30-Minute Student Task

Rule-Based Python Chatbot | Student Submission

1. Objective

The aim is to build a simple rule-based conversational chatbot in Python. It should keep interacting with the user, recognize different messages, give suitable replies and allow the user to exit.

2. Python Program

```
print("Welcome to my chatbot!")

name = input("Bot: What is your name? ").strip()

print(f"Bot: Nice to meet you, {name}!")
print("Bot: You can ask me about Python, weather, or studying.")
print("Bot: Type 'bye' or 'exit' to stop.")

while True:
    message = input(f"{name}: ").lower().strip()

    if message == "bye" or message == "exit":
        print(f"Bot: Goodbye, {name}! Have a great day.")
        break

    elif "hello" in message or "hi" in message or "hey" in message:
        print(f"Bot: Hello, {name}! How can I help you?")

    elif "python" in message:
        print("Bot: Python is a beginner-friendly programming language "
              "used in data science, AI, web development and automation.")

    elif "weather" in message:
        print("Bot: I cannot check live weather, but you can use a weather service "
              "to see the current conditions.")

    elif "study" in message or "exam" in message:
        print("Bot: Make a small study plan, practice questions and review mistakes.")

    else:
        print("Bot: I am still learning. Try Python, weather or studying.")
```

3. Requirements Covered

Requirement	How it is covered
Ask name first	input() asks for the user's name before the chat.
Continuous conversation	while True keeps the program running.
Clean input	.lower().strip() is used.
Greetings	hello, hi and hey are recognized.
Different topics	Python, weather and study/exam are handled.
Decision making	if / elif / else is used.
Exit	bye or exit calls break.
Name in response	The chatbot uses the user's name.
Unknown input	else gives a useful fallback response.
Flexible matching	The in operator recognizes phrases such as hello bot.

4. Testing

Input	Expected result
hello	Greeting response
HELLO	Greeting response after lower()
hello bot	Greeting using in
hi	Greeting response
What is Python?	Python information
How is the weather?	Weather response
How should I study?	Study advice
something random	Unknown-message response
bye / exit	Goodbye and program stops

5. Sample Run

```
Bot: What is your name? Jyothi
Bot: Nice to meet you, Jyothi!
Jyothi: HELLO
Bot: Hello, Jyothi! How can I help you?
Jyothi: tell me about python
Bot: Python is a beginner-friendly programming language used in data science, AI, web development and automation.
Jyothi: bye
Bot: Goodbye, Jyothi! Have a great day.
```

6. Result

The chatbot was successfully designed using basic Python concepts. It follows the required flow: INPUT → PROCESS → RESPOND → REPEAT → EXIT.